Inorganic and Materials Cumulative Exam
February 9, 2018

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. (a) (15pts) Using a molecular orbital diagram, describe the electronic structure of HF (F 1s orbital potential = -40.2eV, F 2s orbital potential = -18.6eV, H 1s orbital potential = -13.6eV).
(b) (5pts) Is the bond polar? Use evidence from your molecular orbital diagram to justify your answer.

2. (a) (15pts) Define linkage isomerism and provide an example of a ligand that can participate in it. Draw diagrams to support your answer.
(b) (5pts) Describe the role that the solvent can play in the bonding of a metal-ligand complex that exhibits linkage isomerism.

3. Do you expect the Ni 2p XPS peak for NiO to be at higher or lower binding energy than Ni metal? Why?

4. (a) (5pts) Draw the molecular orbital diagram of N_2.
(b) (10pts) Calculate the bond order of N_2, N_2^+, and N_2^{2+}.
(c) (5pts) Of the three dinitrogen molecules in (b), which has the shortest bond length?

5. (a) (12pts) What is the point group of NH_3?
(b) (8pts) Using the appropriate character table (included), how many IR-active modes does NH_3 have?
(b) (8pts) Using the appropriate character table (included), how many Raman-active modes does NH_3 have?

6. (a) (10pts) What is the difference between a vacancy defect and an interstitial defect in a crystal lattice?
(b) (10pts) What is the difference between an edge dislocation and a screw dislocation in a crystal lattice?

7. (a) (12pts) In the context of mixed-metal solid solutions, what does it mean for the components to be immiscible?
(b) (8pts) Explain how two metals may be immiscible in a bulk material, yet miscible in a nanoparticle formulation.

8. (a) What is an exciton?
(b) Excitons can exist in semiconductors and in polymers, but their lifetimes in polymers are orders of magnitude longer. Why is this?

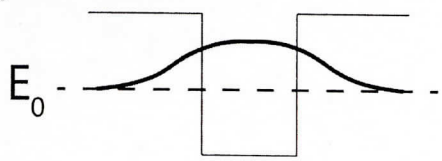
9. (a) (12pts) Use band diagrams to describe the construction and operation of a semiconductor diode.

(b) (8pts) Sketch the current-voltage relationship of a semiconductor and label the regions of forward and reverse bias.

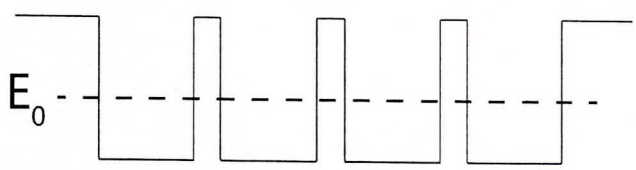
10. A simple derivation of one-dimensional band theory involves building exceedingly longer chains of identical potential wells. In the figure below (a) shows the ground state, s-like electronic wavefunction of a single finite square well potential and its ground state energy E_0. Using this graphical result, draw the unique 'linear combinations of atomic orbitals' that represent the molecular orbitals for the four finite-square well chain (b). Plot the energy of each combination with respect to E_0.

(Figure for question #10)

(a)



(b)



$$\# reps = \frac{1}{h}\sum(g * \chi_r * \chi_i)$$

h = total operations in a group

g = number of operations in a class

χ_r = the character of the reducible representation

χ_i= the character of the irreducible representation

The transformation matrix for a C_n rotation around the z axis is

$$\begin{bmatrix} cos\,\theta & -sin\theta & 0 \\ sin\theta & cos\,\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$cos\frac{2\pi}{1} = 1, sin\frac{2\pi}{1} = 0$$

$$cos\frac{2\pi}{2} = -1, sin\frac{2\pi}{2} = 0$$

$$cos\frac{2\pi}{3} = -0.5, sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$$

$$cos\frac{2\pi}{4} = 0, sin\frac{2\pi}{4} = 1$$

$$cos\frac{2\pi}{5} = \frac{1}{4}(-1+\sqrt{5}), sin\frac{2\pi}{5} = \frac{1}{4}\sqrt{10+2\sqrt{5}}$$

$$cos\frac{2\pi}{6} = 0.5, sin\frac{2\pi}{6} = \frac{\sqrt{3}}{2}$$

Character Tables

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

C_3	E	C_3	$(C_3)^2$		
A	1	1	1	z, R_z	x^2+y^2, z^2
E	1 1	e e^*	e^* e	x+iy; R_x+iR_y x-iy; R_x-iR_y	(x^2-y^2, xy) (yz, xz)

e = exp(2πi/3)

C_4	E	C_4	C_2	$(C_4)^3$		
A	1	1	1	1	z, Rz	x^2+y^2, z^2
B	1	-1	1	-1		x^2-y^2, xy
E	1 1	i -i	-1 -1	-i i	x+iy; R_x+iR_y x-iy; R_x-iR_y	(yz, xz)

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	(x, y) (R_x, R_y)	(x^2-y^2, xy) (xz, yz)

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		x^2-y^2
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	(x, y) (R_x, R_y)	(xz, yz)

C_{6v}	E	$2C_6$	$2C_3$	C_2	$3\sigma_v$	$3\sigma_d$		
A_1	1	1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	1	-1	-1	Rz	
B_1	1	-1	1	-1	1	-1		
B_2	1	-1	1	-1	-1	1		
E_1	2	1	-1	-2	0	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	=1	=1	2	0	0		$(x^2-y^2$, xy)

C_{5v}	E	$2C_5$	$2(C_5)^2$	$5\sigma_v$		
A_1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	-1	Rz	
E_1	2	2cos(2π/5)	2cos(4π/5)	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	2cos(4π/5)	2cos(2π/5)	0		(x2-y2, xy)

D_{2h}	E	C_2 (z)	C_2 (y)	C_2 (x)	i	σ (xy)	σ (xz)	σ (yz)		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	R_z	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	R_y	xz
B_{3g}	1	-1	-1	1	1	-1	-1	1	R_x	yz
A_u	1	1	1	1	-1	-1	-1	-1		
B_{1u}	1	1	-1	-1	-1	-1	1	1	z	
B_{2u}	1	-1	1	-1	-1	1	-1	1	y	
B_{3u}	1	-1	-1	1	-1	1	1	-1	x	

D_{3h}	E	$2C_3$	$3C'_2$	σ_h	$2S_3$	$3\sigma_v$		
A'_1	1	1	1	1	1	1		x^2+y^2, z^2
A'_2	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A''_1	1	1	1	-1	-1	-1		
A''_2	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C'_2$	$2C''_2$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	-1	-1	1	1	1	-1	-1	Rz	
B_{1g}	1	-1	1	1	-1	1	-1	1	1	-1		x^2-y^2
B_{2g}	1	-1	1	-1	1	1	-1	1	-1	1		xy
E_g	2	0	-2	0	0	2	0	-2	0	0	(Rx, Ry)	(xz, yz)
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1		
A_{2u}	1	1	1	-1	-1	-1	-1	-1	1	1	z	
B_{1u}	1	-1	1	1	-1	-1	1	-1	-1	1		
B_{2u}	1	-1	1	-1	1	-1	1	-1	1	-1		
E_u	2	0	-2	0	0	-2	0	2	0	0	(x, y)	

T_d	**E**	**$8C_3$**	**$3C_2$**	**$6S_4$**	**$6\sigma_d$**		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2-x^2-y^2, x^2-y^2)$
T_1	3	0	-1	1	-1	(R_x, R_y, R_z)	
T_2	3	0	-1	-1	1	(x, y, z)	(xy, xz, yz)